

CE 773: Geometric Design and Analysis of High-Speed Roadways

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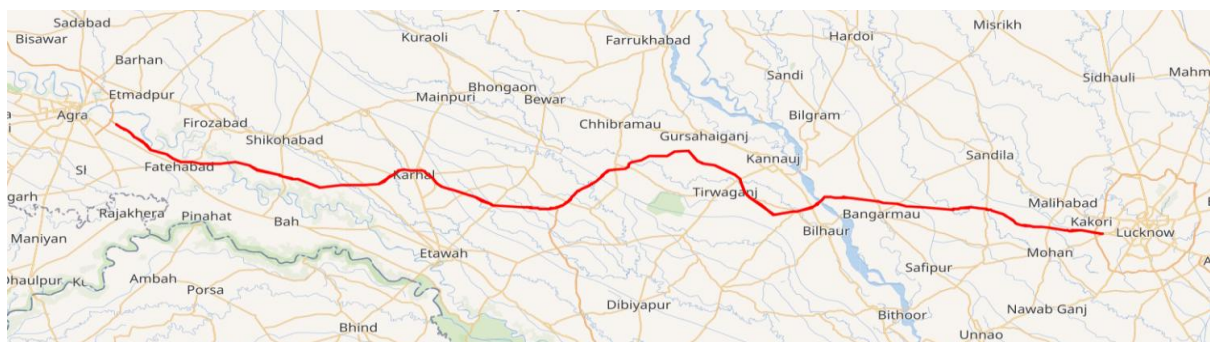
Assignment 1

Review any significant highway development project implemented in the last decade and describe the different design concepts applied.

We are analyzing the **Agra–Lucknow Expressway**, a 302.22 km-long access-controlled, six-lane expressway designed to revolutionize connectivity between two major cities in Uttar Pradesh. This brief review evaluates its geometric design concepts and how it addresses critical challenges in transportation, safety, and sustainability.

Agra–Lucknow Expressway

The Agra–Lucknow Expressway was developed to tackle transportation issues in Uttar Pradesh, significantly reducing travel time from over **6 hours to 3.5 hours between the two cities**. It addresses congestion and safety on **NH-19**, enhancing connectivity between major economic hubs and integrating backward districts **like Kannauj, Auraiya, and Unnao**. It is designed as a greenfield project and features modern amenities like Intelligent Transport Systems and sustainable practices such as rainwater harvesting. The expressway is a crucial catalyst for economic growth, trade, tourism, and regional investments.



Project Overview

Feature	Details
Length	302.22 km
Lanes	6 Lanes, expandable to 8 Lanes
Construction Period	January 2015 to November 2016
Completion Time	22 months
Inauguration Date	21 November 2016
Travel Time Reduction	From 6 hours to approximately 3.5 hours between Agra and Lucknow
Starting Point	Agra Inner Ring Road
Ending Point	State Highway 40 in Lucknow
Districts Covered	Agra, Firozabad, Mainpuri, Etawah, Auraiya, Kannauj, Kanpur Nagar, Hardoi, Unnao, Lucknow

Cost Overview:

Aspect	Details
Initial Estimated Cost	₹15,000 crore
Final Completion Cost	₹13,200 crore
Cost Savings	₹1,800 crore

Major Stakeholder:

Stakeholders	Role
Government of Uttar Pradesh	Project financing and oversight
Uttar Pradesh Expressways Industrial Development Authority (UPEIDA)	Project implementation and management
Construction Contractors	Various firms responsible for different sections of the expressway
Local Communities	Residents of districts through which the expressway passes benefit from improved connectivity and economic opportunities

Different Design Concepts align with the Agra-Lucknow Expressway Project:

1. Greenfield Alignment Design

- The Agra–Lucknow Expressway was built to solve problems with the **busy NH-19 road**, which faced heavy traffic, delays, and safety concerns.
- Expanding the old road would have needed much land in crowded areas, causing more displacement and higher costs.
- The new expressway takes a shorter route around cities, allowing faster and smoother travel. Advanced technologies, **like GIS and LiDAR**, were used.

2. Complete Street Concepts

The Agra–Lucknow Expressway does **not incorporate complete street concepts** to accommodate all users, including pedestrians, cyclists, motorists, and transit vehicles.

The expressway is designed only for fast-moving vehicles and doesn't have bike lanes or sidewalks for non-motorized users. It's focused on high-speed travel and doesn't support mixed-use transportation. However, there are side roads for local access, allowing residents to reach their destinations.



3. Median and Roadside Safety Design

- The expressway has **wide medians and crash barriers** to keep everyone safe.
- The medians are **10–12 meters wide**, helping to stop head-on crashes, while crash barriers make accidents less severe.
- The **landscaped medians** also look nice and help the environment, balancing safety with beauty.

4. Context-Sensitive Solutions (CSS)

- The expressway aligns well with **context-sensitive solutions** by addressing the region's specific needs while minimizing environmental and social impact.
- The alignment was carefully planned to avoid **urban centers, reducing disruptions in densely populated areas**.
- Additionally, the project integrates **green infrastructure, such as green belts** along the expressway, noise barriers to reduce sound pollution, and rainwater harvesting pits every **500 meters** to address water management.

These measures ensure the project is sensitive to the environment and the communities it traverses, making it a sustainable and inclusive infrastructure project.

5. Performance-Based Design

- The expressway's **performance-based design** ensures safety and operational efficiency.
- **Grade-separated intersections** remove level crossings, improving traffic flow and reducing delays.
- Flyovers and underpasses are placed at key crossroads, while **trumpet and cloverleaf designs make interchanges more efficient**. These setups enhance connectivity and road safety by minimizing conflict points.
- Including **Intelligent Transport Systems (ITS)**, such as surveillance cameras, electronic call boxes, digital message boards, and incident detection systems, enhances safety and traffic management.

These elements ensure the expressway performs reliably under varied conditions, reducing travel time and improving safety standards.

6. Practical Design

The expressway adopts a **practical design approach**, focusing on essential features to balance cost and functionality. **For example**, including limited interchanges for access control and service roads for local traffic ensures smooth operation without unnecessary complexity. Additionally, crucial safety measures, such as **rumble strips and reflective signages**, enhance safety while keeping construction costs in check. This approach enabled the expressway to be completed within budget while meeting high operational standards.

Emergency Airstrip Integration

- A standout feature of the expressway is its **3.3 km-long emergency airstrip** near Unnao, designed to support disaster management and defense operations.
- Before this project, India lacked infrastructure capable of handling emergency fighter jet landings on highways.
- The airstrip supports the **Indian Air Force (IAF)** and disaster relief operations, ensuring dual-purpose utility. This innovation not only serves transportation needs but also **strengthens national security infrastructure**.

7. Design Matrix Approach

The **design matrix approach** was **not explicitly applied** during the planning and construction phases of the expressway. While this method typically involves choosing between basic, modified, or complete design standards based on project goals, the expressway adheres to high-speed geometric standards suitable for its purpose. Despite lacking a formal design matrix, the project's meticulous planning ensured alignment with modern expressway requirements.

8. Safe System Approach

- The expressway prioritizes safety with well-designed curves, wide medians, crash barriers, and **a 120 km/h speed limit**.
- The use of ITS enhances safety by ensuring prompt incident detection and response.
- Use of **Green Energy (Solar)** for all illumination purposes.
- **Service Road**, staggered on both sides of the Expressway except at Major bridges, RoBs.
- Patrolling vehicles and GPS-based ambulance services ensure rapid assistance within **4–11 minutes, minimizing the impact of accidents**.

9. Travel Time Reliability Approach

The travel time reliability approach on the expressway has improved consistency and predictability by eliminating traffic signals, restricting access, and integrating real-time traffic monitoring. **This has reduced travel time between Agra and Lucknow to about 3.5 hours, benefiting commuters and freight operators.**

10. Value Engineering

- The expressway successfully utilized **value engineering** to optimize costs without compromising quality.
- Through efficient land acquisition, streamlined construction processes, and practical design practices, the project achieved cost savings of ₹1,800 crore, **reducing the total expenditure from the initial estimate of ₹15,000 crore to ₹13,200 crore.**

This demonstrates how resourceful planning can deliver high-quality infrastructure within budget constraints.

11. Designing of 3R Projects

The Agra–Lucknow Expressway is a greenfield project built from scratch and **does not fall under 3R projects (Resurfacing, Restoration, Rehabilitation)**. Unlike 3R projects that upgrade existing infrastructure, it is a new access-controlled highway designed for high-speed connectivity without links to older roads.

12. Performance-Based Practical Design (PBPD)

The expressway is designed for good performance while keeping costs down. It starts with six lanes but can be **expanded to eight** if needed. It includes innovative technology for traffic management and green spaces for sustainability. This design focuses on safety, usability, and budget, ensuring it can adapt to future needs.

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Assignment 2

1. In a rural median-divided four-lane highway development project, you must find the design traffic volume. The hourly traffic volume (in PCU) of 50 days per calendar year for one direction is provided in the spreadsheet “Assignment-2.xlsx”. Choose data corresponding to your roll number and estimate the ADT and 30th-hour Design Volume. Step by step, show the calculations.

Step Wise Solution:

Given Data:

- Rural Median – Divided Four-Lane Highway
- hourly traffic volume (in PCU) **of 50 days** per calendar year for **one direction**

Objective:

- Estimate the ADT
- 30th-hour Design Volume

- i. Let's start with the estimation of the ADT:

ADT = Average Daily Traffic = The ADT is the total volume during a given period (in whole days), greater than one day and less than one year, divided by the number of days in that period.

Here, Time period = 50 Days

The sum of Daily Traffic (for 50 Days) = 1638192

Thus ADT = $1638192/50 = 32763.84$ **PCU/day/Lane**

Inference: The average number of vehicles passing the section per day over the observed period was 32763 PCU per day per lane.

- ii. Estimation of 30th Hour Design Volume:

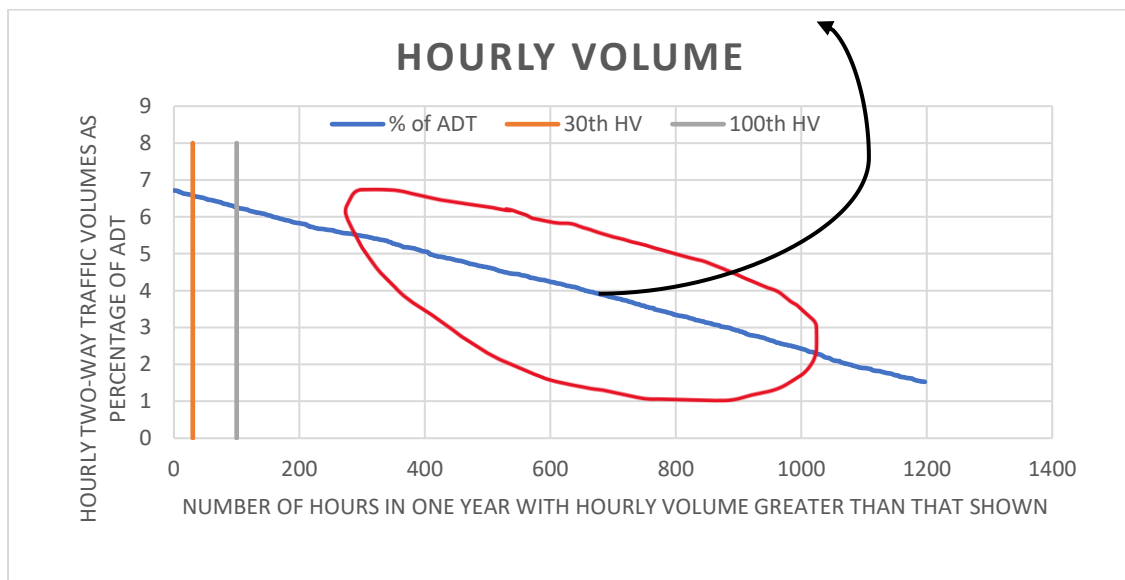
Step 1. Combine all the traffic Data for 50 days in one column. We will get ($50 \times 24 = 1200$) hr rows.

Step 2: Sort them in descending order, arranging the data from the highest to the lowest traffic volume.

Step 3: Identify the 30th Highest Value: The 30th highest value corresponds to the traffic volume exceeding 29 times.

Result: After Sorting the 30th Hour Design Volume Comes = **2157 PCU/hr/Lane**

This comes straight line as we have the data set of only 50 days, not 1 year. This will affect the **Design Volume Consideration**



➔ Since there is no sizeable seasonal variation, the 30th highest volume = 15% of ADT

- The operating speed of the highway in question 1 is 100 kmph. Consider the estimated design volume to review the design improvement required in the next 10 years. The expected LOS at any point should not fall below “C.” For any improvement to be implemented, it takes about 3 years for design, ROW acquisition, and construction. Step by step, it shows the evaluation and decision-making process.

Step Wise Solution:

Given Data:

- Operating Speed = 100 kmph, and based on this, we will select the Capacity Value for four and six lanes.

Base capacity of multilane highways



Operating speed (kmph)	Capacity (PCU/hr/direction) for 4-lane divided highways	Capacity (PCU/hr/direction) for 6-lane divided highways
70	3640	5500
80	3940	5930
90	4240	6360
100	4540	6790

Thus, Capacity for 4 Lanes = 4540 PCU per hr per 2 Lane and

for 6 Lane Divided Highways = 6790 PCU per hour per 3 lane

- Growth Rate = 5% [According to the **Indian Roads Congress (IRC)**]

Given Constraints:

- Expected LOS at any point should not fall below “C.”

LOS & Service Volume: Four-lane divided inter-urban roads



- Indo-HCM 2017

LOS	Density (PCU/km/direction)	V/C Ratio	Service volume (PCU/day)
A	≤ 18	≤ 0.20	< 18000
B	19 – 27	0.21 – 0.30	18001 – 27000
C	28 – 45	0.31 – 0.50	27001 – 45000
D	46 – 64	0.51 – 0.70	45001 – 63000
E	65 – 90	0.71 – 1.0	13001 – 90000
F	> 90	> 1.0	> 90000

- Suggested threshold value to convert four-lane divided road to six-lane divided road: 22500 PCUs/day at LOS B

- It takes about 3 years to design, acquire ROW, and construct.

Objective:

Consider the estimated design volume to review the design improvement required in the next 10 years.

Step 1: Analyze the present scenario.

What is the LOS based on present design volume and Capacity, and what will be the LOS after 10 years?

- **Present Design Volume = 2157 PCU/hr/Lane**
- **Present Capacity = 4540 PCU/hr/Lane**

The formula for future volume with growth rate is:

$$\text{Future Volume} = \text{Current Volume} \times (1 + \text{Growth Rate})^{\text{Years}}$$

Given a **growth rate of 5%**:

$$\text{Future Volume} = 2157 \times (1 + 0.05)^{10} = 3513.53$$

		Cur rent year	1	2	3	4	5	6	7	8	9	10 th Yea r
	Desi gn Volu me	2157	2264. 85	2378. 0925	24 97	262 1.85	275 2.94	289 0.59	303 5.12	318 6.87	334 6.21	351 3.53
4 Lan e divi ded	Cap acity	4540	4540	4540	45 40	454 0	454 0	454 0	454 0	454 0	454 0	454 0
	v/c	0.47 511	0.498 8656	0.523 8089	0. 55	0.57 75	0.60 637	0.63 669	0.66 853	0.70 195	0.73 705	0.77 39
	LOS	C	C	D	D	D	D	D	D	E	E	E

- ➔ The 4-lane highway consistently maintains a Level of Service (LOS) at C until year 2, after which it drops to LOS 'D' and LOS 'E.'
- ➔ From here, we can observe that LOS starts dropping after 1st year, so if we consider this 30th hourly Design volume, we will not get enough time to start the construction. We need 3 years to complete the construction as per the questions.

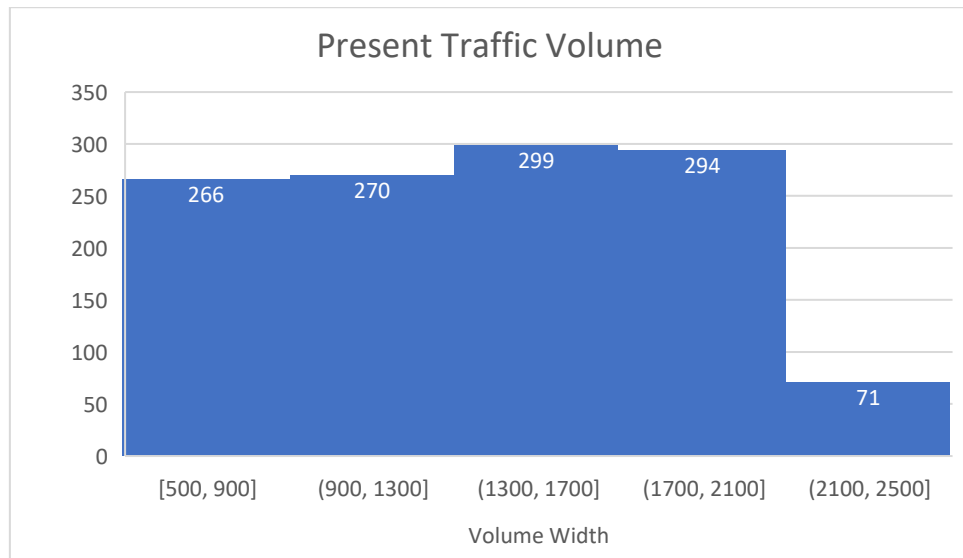
Thus, we must change this present Design Volume and select the suitable design hourly volume.

Step 2: Selecting the suitable Design Volume and Justification:

According to IRC (Design Hourly Volume for Rural Four-Lane Highways):

*“Design typically assumes moderate traffic growth for rural four-lane highways over the next 10-20 years. The **95th percentile volume** is sometimes used to account for more extreme traffic conditions that may occur in the future.”*

Here, if we **analyze the traffic** of this one lane, we observe:



OBSERVATION:

- Most traffic volume (266, 270, 299, 294) falls within the lower ranges, **between 500 and 2100 vehicles**.
- The **[2100, 2500]** range has the **least volume**, with only **71 vehicles**, indicating that high traffic volumes in this range are rare.

The data indicates that the roadway experiences **moderate traffic conditions**, characterized by infrequent occurrences of exceptionally high traffic volumes, typically ranging from 2,100 to 2,500 vehicles.

Conclusion:

Thus, we will take the **85th Percentile Volume** as our Estimated Design volume.

Step 3: Calculation of 85th Percentile Volume

To find the **85th percentile rank** in a sorted list(In Ascending Order) of **1200** values:

- **Percentile rank = $0.85 \times (N)$**
- Here, We have **1200 hourly volumes** (from 50 days $\times$ 24 hours); I would use the formula to find the rank and then pick the value in the sorted list.
 - Percentile Rank = $0.85 \times 1200 = 1020^{\text{th}}$ Value

From the Data:

Percentile :	Position(th)	Design Volume (PCU per hr per 2 lane)
97.5th Percentile	1164	2157
95th Percentile	1140	2115
85th Percentile	1020	1935

Step 4: Calculation of Traffic Volume and growth on the 85th Percentile Volume

		Year -->	1	2	3	4	5	6	7	8	9	10
	Design Volume	1935	2031.75	2133.3375	2240	2352	2469.6	2593.09	2722.74	2858.88	3001.82	3151.91
4 Lane divided	Capacity	4540	4540	4540	4540	4540	4540	4540	4540	4540	4540	4540
	v/c	0.426211	0.447522	0.4698981	0.49339	0.51806	0.54397	0.57116	0.59972	0.62971	0.66119	0.69425
	LOS	C	C	C	C	D	D	D	D	D	D	D
6 Lane divided	Capacity	0	0	0	0	6790	6790	6790	6790	6790	6790	6790
	v/c	0	0	0	0	0.34639	0.36371	0.3819	0.40099	0.42104	0.44209	0.4642
	LOS	0	0	0	0	C	C	C	C	C	C	C

Overall Result:

- ➔ The 6-lane highway starts from year 4, with a capacity of 6790 vehicles. The v/c ratio for the 6-lane configuration remains well below the threshold for LOS D, stabilizing at around 0.364 to 0.464 throughout the years.
- ➔ By expanding(Starting the construction) from 4 to 6 lanes starting from the present year, the road would maintain LOS C in the initial years and avoid the drop to LOS D in year 4.

LOS & Service Volume: Six-lane divided inter-urban roads



- Indo-HCM 2017

LOS	Density (PCU/km/direction)	V/C Ratio	Service volume (PCU/day)
A	≤ 27	≤ 0.20	< 27000
B	28 – 41	0.21 – 0.30	27001 – 41000
C	42 – 68	0.31 – 0.50	41001 – 68000
D	69 – 95	0.51 – 0.70	68001 – 95000
E	96 – 136	0.71 – 1.0	95001 – 136000
F	> 136	> 1.0	> 1360000

- Suggested threshold value to convert six-lane divided road to eight-lane divided road: 34000 PCUs/day at LOS B

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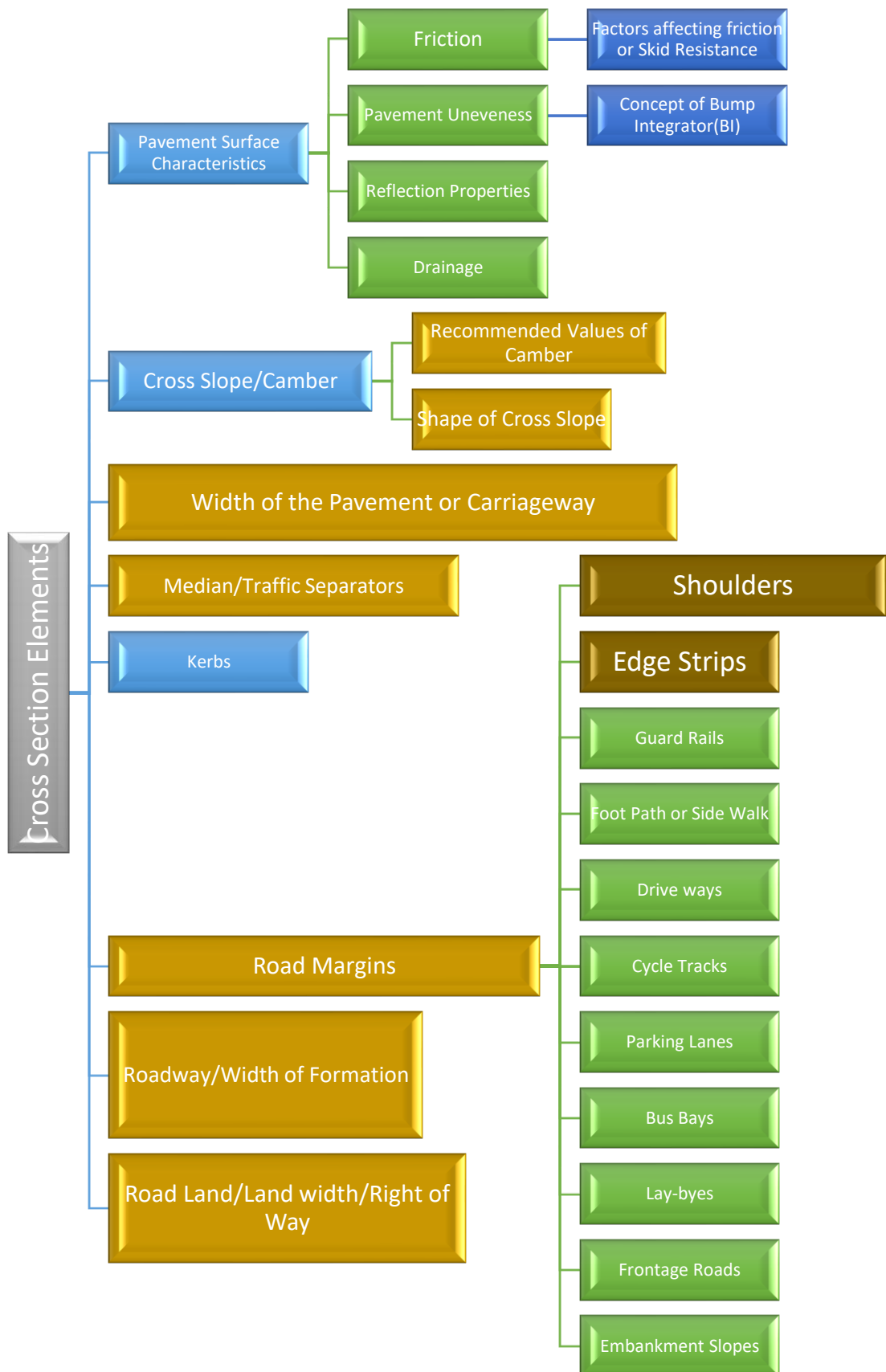
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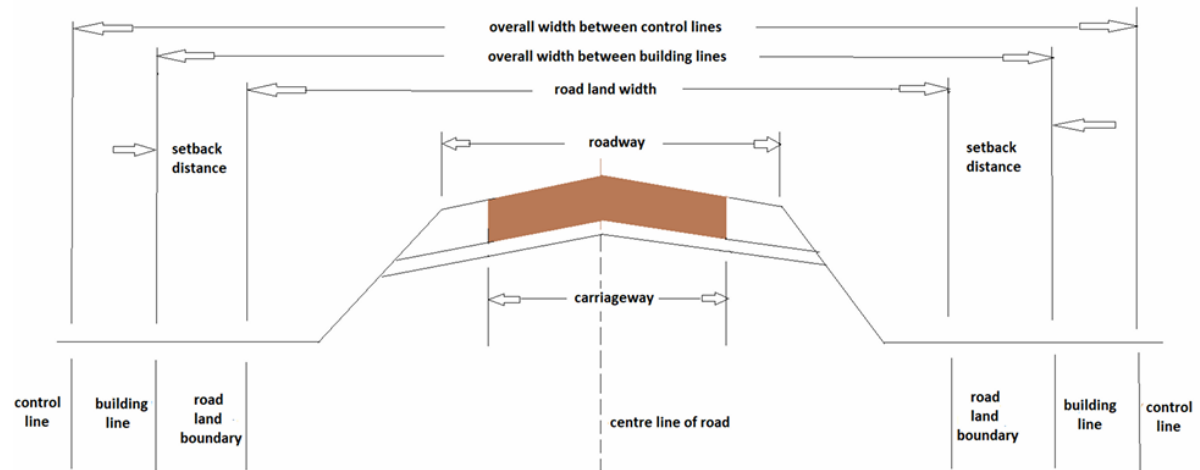
Assignment 3

1. Develop the cross-section of a **four-lane expressway** for a **tangent section**. This expressway is located in **the open country** of the **northeast region of India**, where the annual rainfall is about 200 cm. Assume appropriate details from the relevant IRC code(s). Choose a **suitable horizontal and vertical scale** to represent the cross-section on graph paper.

Solution: Here, Typical Cross-Section Elements are:



We needed only those highlighted in yellow to develop the cross-section of a four-lane expressway for a tangent section.



Design Considerations:

1. **Location:** Open country in North-East India, with an annual rainfall of 200 cm.
2. **Road Classification:** Expressway (4-lane).
3. **Pavement Surface Type:** IRC recommends using a bituminous surface for high-speed roads.
4. **Lane Width:** According to the IRC guidelines for expressways in plain and rolling terrain, the lane width is 3.75 m per lane.
 - For a four-lane highway, the **total carriageway width** is $4 \times 3.5 \text{ m} = 14 \text{ m}$.

Carriageway width in urban roads (IRC:86-2018)

Description	Width (m)
Single lane with raised kerbs	3.50
2-lane with raised kerbs	7.00
4-lanes with raised kerbs	14.0
6-lanes with raised kerbs	21.0
8-lanes without kerbs	28.0

- For access roads to residential areas, a lower lane width of 3 m is permissible
- Minimum width of urban road without kerb shall be 5.5 m including allowances for a stalled vehicle and pedestrian movement.

5. Shoulder Width:

- **Paved Shoulder:** A minimum shoulder width of 3 m is recommended on both sides.

Terrain	Paved shoulder (m)
Plain	3.0
Rolling	3.0
Mountainous	2.5

- **Earthen Shoulder:** A minimum shoulder width of 1.5 m is recommended on both sides.

$$\text{Total shoulder width} = 3 + 1.5 = 4.5 \text{ m}$$

6. **Median Width:** Here, we will use a 10 m median.

Median width in expressway

(MoRTH: Guidelines for expressways)



Considerations	Recommended width	
	Minimum	Desirable
Without median barrier	10 m	15 m
With median barrier (depends on barrier type)	-	4.5 m
With median barrier (to accommodate structure/pier)	-	8.0 m
For future widening	12 m	15.5 m

- Depressed median with crossfall of 1 in 8 (1 in 6 on narrow median) is recommended. The crossfall can be 1 in 4 to meet the highest amount of overlay.
- Raised median with mountable kerbs may be provided in curves with superelevation

7. **Service Lane:** Here, we are taking the service lane of 5m.
8. **Crossfall or Camber:** For a bituminous surface in regions with heavy rainfall (≥ 100 cm), the camber is **2.5%**.

Camber in Expressways

(MoRTH: Guidelines for expressways)



Cross-sectional elements	Surface	Rainfall	
		100 cm or more	Less than 100 cm
Pavement, edge strip and pavement shoulder	Bituminous	2.5%	2.0%
	Concrete	2.5%	2.0%

9. **Side Slopes:** For expressways, the side slopes of embankments are recommended to be **4H:1V** or flatter for moderate-height embankments.
10. **Right of Way:** The **Right of Way (RoW)** refers to the total land width required to accommodate:
- Roadway (carriageway, shoulders, median)
 - Side drains
 - Service roads

- Utilities

For a **four-lane expressway** in **open country** with a design speed of 100 km/h, the **RoW** is typically designed as follows based on IRC guidelines:

- For **expressways**, the RoW generally ranges from **90 m** in open terrain (plain or rolling).

Right of Way of Multi-Lane highway



Type	ROW	Ref.
2-lane highway	Should accommodate roadway, side drains, service road, tree plantation, utilities, etc. Desirable width should meet IRC:73 and IRC:86	IRC:SP:73-2007
4-lane highway	Minimum 60 m	IRC:SP:84-2019
6-lane highway	Minimum 60 m	IRC:SP:87-2010
Expressway	Plain and rolling terrain: 90 m Mountainous same level: 60 m Mountainous split level: 30 m	MoRTH: Guideline for Expressways

11. Building and Control Lines

Building and Control Lines

(MoRTH: Guidelines for expressways)



Terrain	Open areas		Built-up areas
	Overall width between building lines (m)	Overall width between control lines (m)	Distance between building line and RoW boundary (m)
Plain	110	130	5
Rolling	110	130	5
Mountainous	70	80	5



REVIEW OF INTERSECTION

- Location: H10 Intersection of IIT Bombay Campus

Assignment 4

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1. Introduction

○ Objective

The objective of this study is to conduct a **geometric design and safety evaluation** of the H10 intersection within the IIT Bombay campus. The review is grounded in the principles of intersection design as taught in the CE773 course: *Geometric Design and Analysis of High-Speed Roadways*.

○ Scope

The evaluation is based on **field observations**, **photographic evidence**, and **intersection design guidelines**, focusing strictly on geometric elements such as sight distance, alignment, pedestrian and cyclist integration, and signage placement.

📌 Location Description

The selected intersection is located near **Hostel 10 (H10)** on the IIT Bombay campus, connecting the **H14 hostel road**, the **H12 hostel approach**, and a minor arm leading toward **CCD and KV School**. The image below shows a view of one approach arm at the intersection.



Figure 1(Fig 1: Taken from the H12 side toward the intersection with visible signs and curvature).

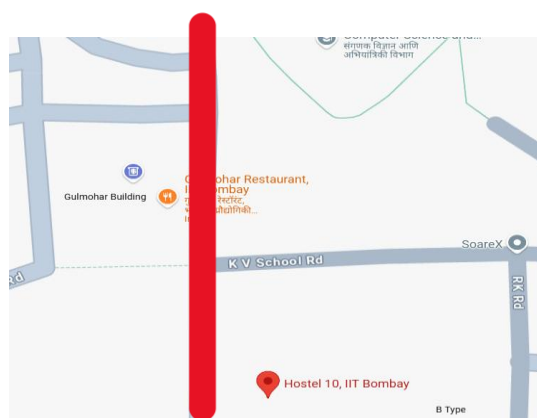


Figure 2Fig 2. Google Map view (Ariel View)

Purpose of the Intersection

The intersection facilitates movement for:

- **Hostel residents** (pedestrians and cyclists),
- **Institutional traffic** (autos, service vehicles),
- And **access to key campus facilities** (IDC, Hostels 10, 12, 14).

It experiences moderate daily traffic as a T-intersection, **with a mixed user group**. This makes it critical to evaluate how well the current geometric design supports **safety**, **clarity**, and **efficiency** for all modes of travel.

2. Existing Layout and Traffic Movement

- Intersection Type

The intersection under study is a **3-legged (T-intersection)**, located near H10 and H12 hostels within the IIT Bombay campus.

- Traffic Control Features

- a. : No yield or stop sign is provided to control the intersection.
- b. The intersection operates in an uncontrolled condition, relying on driver judgment and road-user negotiation.

- Lane and Movement Characteristics

- a). Each leg of the intersection consists of **two lanes with two-way traffic permitted**.
- b). All turning movements are allowed, including:
 - Right-turns from the minor leg (H12) onto the major road.
 - Left-turns into and from all approaches.

- Traffic Composition

The intersection experiences a **heterogeneous mix of users**, including:

- **Passenger vehicles:** cars, two-wheelers, and autorickshaws
- **Non-motorized users:** pedestrians and cyclists
- **Service and delivery vehicles:** including light and heavy commercial vehicles (LCVs and HCVs) accessing nearby facilities like the SAC and hostels

- Geometric Challenges from Movement

Due to the absence of channelization and traffic control, the intersection currently functions with *limited guidance for right-of-way*, increasing the potential for:

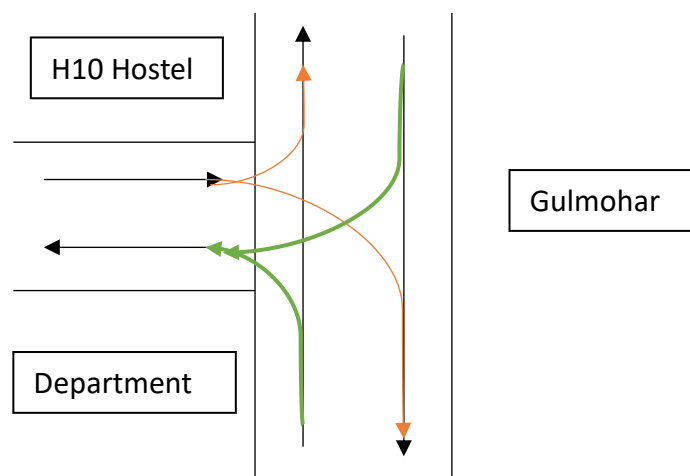
- Crossing conflicts between right-turning vehicles and oncoming traffic.
- Merging and diverging conflicts at the T-junction without geometric separation.
- Pedestrian-vehicle confusion due to the lack of crosswalks and pedestrian refuge islands.

3. Conflict Point Analysis

- Based on the turning and through movement possibilities at this **three-legged unsignalized intersection**, the following conflict points are identified:

Conflict Type	Count
Crossing Conflicts	3
Merging Conflicts	4
Diverging Conflicts	4
Total Conflicts	11

Thus, there are 11 conflict points.



- Observation of Conflict Risk

- Obstructed Sight Triangle



Figure 3: Block Visibility

- Right-turning vehicles from the H12 approach** face visibility issues due to vegetation, or parked vehicles, increasing the likelihood of side-impact conflicts at crossing points.

- **No Pedestrian Crossings**



Figure 4: No Zebra-Crossing near intersection

- Without marked crossings or refuge islands, **pedestrians cross informally**, creating **uncontrolled pedestrian-vehicle conflict zones** that overlap with vehicle paths.

- **Lack of Regulatory Signs or Markings:**

- The **absence of Stop or Give Way signs and markings** means drivers must guess priority, leading to **unregulated merging and diverging behavior**—especially during **peak hostel hours**.

4. Skew Angle and Intersection Geometry

- **Measured Intersection Angle**

Based on the layout and road alignment visible in the field map and physical inspection, the **angle of intersection** formed at the H10 intersection is estimated to be approximately **85° to 90°**.

- **Interpretation:**

- **Recommended angle for direct crossings:**
75°–105° (as per standard design guidelines)
- **Your intersection:**
Falls within this range → classified as a **“Direct” intersection geometry.**

5. Sight Triangle Evaluation

- **Intersection Control Type: Case A – No Control**

- There are **no stop signs, yield signs, or signals** on any approach.
- Drivers must rely entirely on visual judgment to identify gaps and cross conflicting streams.

6. Problem Identification

1. **Lack of Dedicated Space for Non-Motorized Users:** No cycle lanes or shared pedestrian–bicycle pathways, leading to conflicts between vulnerable users and motorized vehicles.
2. **Inadequate and Obstructed Traffic Signage:** Traffic signs are either too small, poorly placed, or hidden by vegetation, reducing their visibility and effectiveness.
3. **Unclear Right-of-Way Due to Missing Regulatory Elements:** No STOP or GIVE WAY signs or pavement markings, resulting in driver confusion and unregulated movement at the intersection.
4. **Non-Standard and Unsafe Pedestrian Infrastructure:** Uneven curbs at bell mouths, non-uniform walkway heights, and speed breakers and footpaths are on the same level.



Figure 5: Same level of speed breaker and footpath.



Figure 6: Missing Pavement Marking

1. **Faded or Missing Pavement:**
Lack of visual guidance and lane definition, especially under low-light or wet conditions.

5. **Sunken or Damaged Kerbs:**



Figure 7: Damaged Kerb

7. Geometric Design Evaluation

Category	Rating
Uniformity	Needs Improvement
Conflict Minimization	Needs Improvement
Safety	Poor
Alignment & Profile	Acceptable

8. Conclusion:

While the intersection geometry (in terms of angle and profile) is acceptable, the **overall intersection design does not align well with principles**:

- Lack of regulatory signage and markings undermines **simplicity and uniformity**
- High conflict points and unmarked user paths raise **safety concerns**
- Absence of pedestrian and cyclist facilities makes it **non-inclusive**

This makes a strong case for **geometric redesign**, particularly through:

- Channelization
- Visibility clearance
- Signage and markings
- Dedicated paths for NMT (Non-Motorized Transport)

9. Proposed Design Improvements

Based on field observations and geometric evaluation of the H10 intersection, the following **targeted safety improvements** are proposed. These aim to reduce conflicts, improve visibility, and enhance user experience.

A. Non-Motorized User Facilities

1. Dedicated Cycle Lanes / Shared Pathways

- Introduce dedicated bicycle lanes or shared pedestrian–cyclist pathways along intersection arms.
- Reduces lateral conflicts between vulnerable users and motorized vehicles.

2. Marked Pedestrian Crosswalks

- Provide zebra crossings on all legs to channel pedestrian movement.

3. Uniform Pedestrian Walkway Height

- Ensure consistent footpath levels across the approaches to eliminate tripping risks.

4. Level Kerbs

- Standardize curb heights near turning zones for safer pedestrian crossing transitions.

B. Speed Control & Driver Behavior Management

5. Speed Breakers / Rumble Strips

- Install on all approaches at appropriate distances to reduce speed and improve driver attention.

6. Rectify Oversized Speed Hump on H12 Arm

- Adjust the height to ensure the walkway remains elevated and the hump provides only mild deceleration.

7. Post Speed Limit Signboards

- Define and enforce safe operating speeds (e.g., 25–30 km/h) using standard campus signboards.

C. Signage, Markings & Priority Control

8. Install Regulatory Signs (STOP / GIVE WAY)

- Place mandatory signs on the H12 approach and other minor arms to assign clear right-of-way priority.

9. Replace Obscured or Small Traffic Signs

- Upgrade to more prominent retro-reflective signs placed at standard eye level and cleared of vegetation.

D. Visibility and Sight Triangle Enhancements

12. Clear Sight Triangle on H12 Arm

- Remove vegetation and visual obstructions within 25–30 m of the driver’s decision point on the minor arm.

E. Geometric Redesign Suggestions

13. Repair or Replace Sunken Median Kerbs

- Use raised medians or plastic channelizers to clearly separate turning traffic and discourage corner-cutting.

14. Close Canara Bank Driveway Entry

- Block vehicular entry from the bank side near the junction. It introduces unexpected diverging paths and increases merge conflicts.

Category	Key Action
Pedestrian/Cyclist Safety	Crosswalks, kerb leveling, shared pathways
Speed Control	Rumble strips, proper hump design, signs
Traffic Control	STOP/GIVE WAY signs & markings
Visibility	Sight triangle clearance, sign upgrades
Geometric Enhancements	Kerb repairs, channelization, driveway closure